

JAVA PROGRAMMING LAB (BCS154)

B.Tech CSE FIRST YEAR

SUBMMITED TO:

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Assignment 1: Hello World, Variables and Arithmetic

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Print text, store values in variables and calculate with arithmetic operators.*

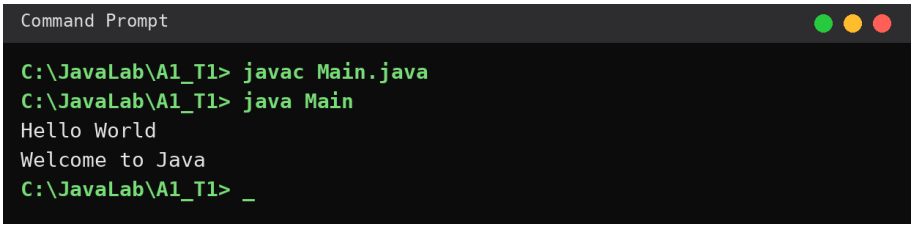
Task 1: Print Hello World on one line. Then print Welcome to Java on the next line.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 1 - Task 1

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World");
        System.out.println("Welcome to Java");
    }
}
```

Output:



Task 2: Create int roll = 101, double percentage = 84.5, char section = 'B' and String name = "Asha". Print each value on its own line, after the labels Roll:, Percentage:, Section: and Name: in that order.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 1 - Task 2

public class Main {
    public static void main(String[] args) {
        int roll = 101;
        double percentage = 84.5;
        char section = 'B';
        String name = "Asha";

        System.out.println("Roll: " + roll);
        System.out.println("Percentage: " + percentage);
        System.out.println("Section: " + section);
        System.out.println("Name: " + name);
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A1_T2> javac Main.java
C:\JavaLab\A1_T2> java Main
Roll: 101
Percentage: 84.5
Section: B
Name: Asha
C:\JavaLab\A1_T2> _
```

Task 3: Create int variables a = 17 and b = 5. Print Sum:, Difference:, Product:, Quotient: and Remainder:, each followed by its result for a + b, a - b, a * b, a / b and a % b.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 1 - Task 3

public class Main {
    public static void main(String[] args) {
        int a = 17, b = 5;

        System.out.println("Sum: " + (a + b));
        System.out.println("Difference: " + (a - b));
        System.out.println("Product: " + (a * b));
        System.out.println("Quotient: " + (a / b));
        System.out.println("Remainder: " + (a % b));
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A1_T3> javac Main.java
C:\JavaLab\A1_T3> java Main
Sum: 22
Difference: 12
Product: 85
Quotient: 3
Remainder: 2
C:\JavaLab\A1_T3> _
```

Task 4: Create a constant named PI with the final keyword and the value 3.14. For a circle with radius 7, print Area: followed by the area and Perimeter: followed by the perimeter, using PI.

Program (Main.java):

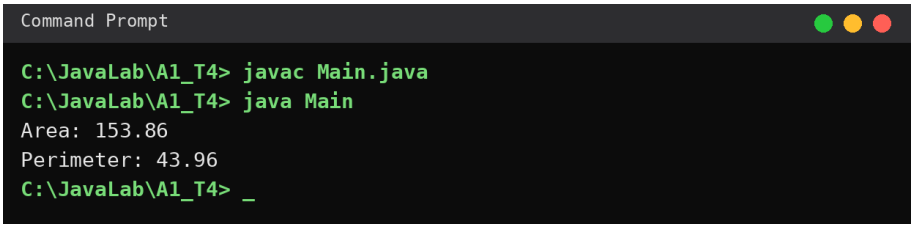
```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 1 - Task 4

public class Main {
    public static void main(String[] args) {
        final double PI = 3.14;
        int radius = 7;

        double area = PI * radius * radius;
        double perimeter = 2 * PI * radius;

        System.out.printf("Area: %.2f%n", area);
        System.out.printf("Perimeter: %.2f%n", perimeter);
    }
}
```

Output:



Assignment 2: Decisions and Loops with Numbers

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Use if, else if and loops to make decisions and repeat steps with whole numbers.*

Task 1: Store 27 in an int named n. Print the value of n followed by is even or is odd, using the % operator and an if-else statement.

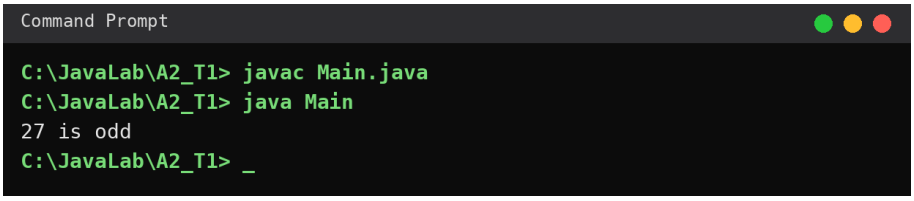
Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 2 - Task 1

public class Main {
    public static void main(String[] args) {
        int n = 27;

        if (n % 2 == 0) {
            System.out.println(n + " is even");
        } else {
            System.out.println(n + " is odd");
        }
    }
}
```

Output:



Task 2: Store 83 in an int named marks. Print Grade: followed by A for 90 or more, B for 75 to 89, and C for less than 75. Use an else-if ladder.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 2 - Task 2

public class Main {
    public static void main(String[] args) {
        int marks = 83;
        char grade;

        if (marks >= 90) {
            grade = 'A';
        } else if (marks >= 75) {
            grade = 'B';
        } else {
            grade = 'C';
        }
        System.out.println("Grade: " + grade);
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A2_T2> javac Main.java
C:\JavaLab\A2_T2> java Main
Grade: B
C:\JavaLab\A2_T2> _
```

Task 3: Store 2024 in an int named year. A leap year is divisible by 4 but not by 100, or divisible by 400. Print Leap year for a leap year, otherwise Not a leap year.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 2 - Task 3

public class Main {
    public static void main(String[] args) {
        int year = 2024;

        if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0) {
            System.out.println("Leap year");
        } else {
            System.out.println("Not a leap year");
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A2_T3> javac Main.java
C:\JavaLab\A2_T3> java Main
Leap year
C:\JavaLab\A2_T3> _
```

Task 4: Use a for loop to find the factorial of 5. Print Factorial of 5 is followed by the value.

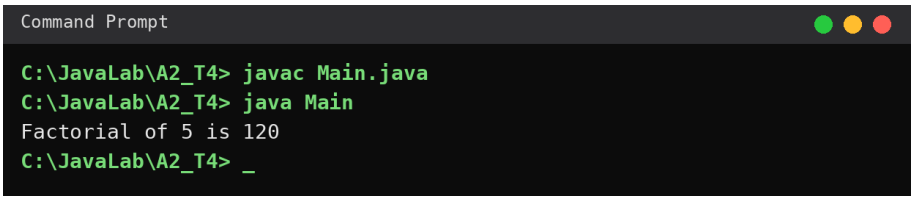
Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 2 - Task 4

public class Main {
    public static void main(String[] args) {
        int fact = 1;

        for (int i = 1; i <= 5; i++) {
            fact = fact * i;
        }
        System.out.println("Factorial of 5 is " + fact);
    }
}
```

Output:



Assignment 3: Star and Number Patterns

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Use a loop inside another loop to print stars and numbers in rows.*

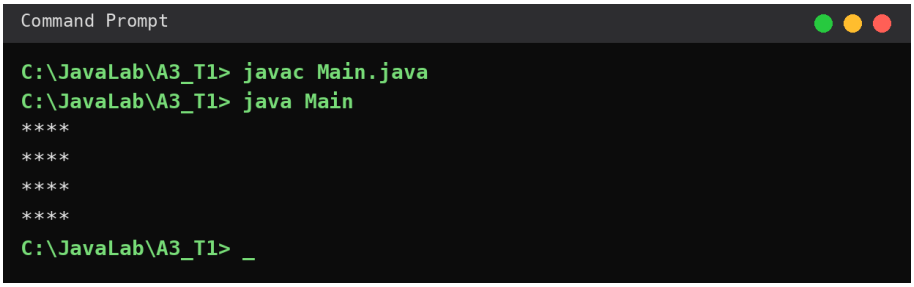
Task 1: Print a square of stars with 4 rows and 4 columns. Use one loop inside another loop.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 3 - Task 1

public class Main {
    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {          // rows
            for (int j = 1; j <= 4; j++) {      // columns
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:



Task 2: Print a right triangle of stars with 4 rows. Row 1 has one star, row 2 has two stars, and so on.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 3 - Task 2

public class Main {
    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A3_T2> javac Main.java
C:\JavaLab\A3_T2> java Main
*
**
***
****
C:\JavaLab\A3_T2> _
```

Task 3: Print a number triangle with 4 rows. Row 1 prints 1. Row 2 prints 1 2. Row 3 prints 1 2 3, and so on. Put a space after each number.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 3 - Task 3

public class Main {
    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A3_T3> javac Main.java
C:\JavaLab\A3_T3> java Main
1
1 2
1 2 3
1 2 3 4
C:\JavaLab\A3_T3> _
```

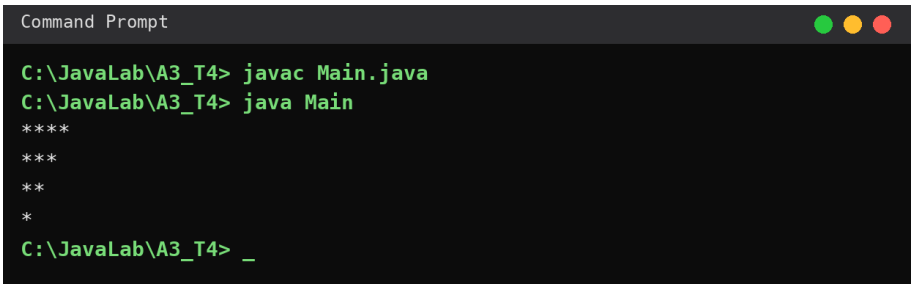
Task 4: Print an upside-down triangle of stars with 4 rows. Row 1 has four stars, row 2 has three stars, and so on, down to one star in row 4.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 3 - Task 4

public class Main {
    public static void main(String[] args) {
        for (int i = 4; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:



Assignment 4: Writing Functions

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Write functions with parameters and return values, and call them from main.*

Task 1: Write a function fact(int n) that returns n factorial, using a loop, and a function nCr(int n, int r) that returns n C r, using fact. In main, print 5C2 = followed by the result of nCr(5, 2).

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 4 - Task 1

public class Main {
    static int fact(int n) {
        int f = 1;
        for (int i = 1; i <= n; i++) {
            f = f * i;
        }
        return f;
    }

    static int nCr(int n, int r) {
        return fact(n) / (fact(r) * fact(n - r));
    }

    public static void main(String[] args) {
        System.out.println("5C2 = " + nCr(5, 2));
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A4_T1> javac Main.java
C:\JavaLab\A4_T1> java Main
5C2 = 10
C:\JavaLab\A4_T1> _
```

Task 2: Write fact(int n) that returns n factorial using a loop, and nPr(int n, int r) that returns n P r using fact. In main, print 5P2 = followed by the result of nPr(5, 2).

Program (Main.java):

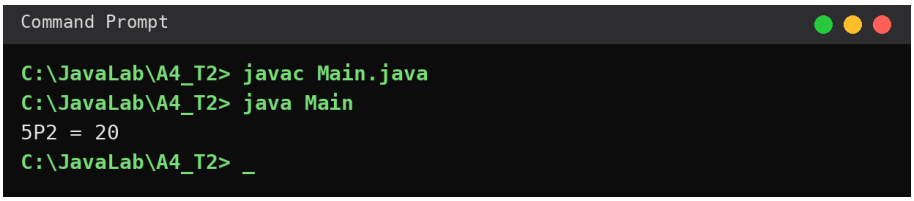
```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 4 - Task 2

public class Main {
    static int fact(int n) {
        int f = 1;
        for (int i = 1; i <= n; i++) {
            f = f * i;
        }
        return f;
    }

    static int nPr(int n, int r) {
        return fact(n) / fact(n - r);
    }

    public static void main(String[] args) {
        System.out.println("5P2 = " + nPr(5, 2));
    }
}
```

Output:



Task 3: Write maxOf(int a, int b) that returns the larger of its two int values. In main, print Larger: followed by the result of maxOf(14, 9).

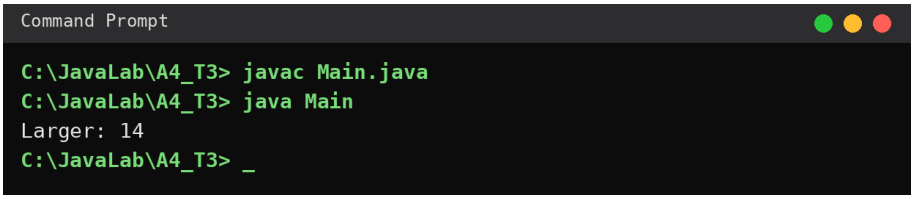
Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 4 - Task 3

public class Main {
    static int maxOf(int a, int b) {
        if (a > b) {
            return a;
        }
        return b;
    }

    public static void main(String[] args) {
        System.out.println("Larger: " + maxOf(14, 9));
    }
}
```

Output:



Assignment 5: Arrays: Traversal and Counting

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Create an array, visit every element with a loop, and add up or count the values.*

Task 1: Create an int array named arr with the values 3, 4 and 5. Use a for loop with arr.length to print one line for each element: the label Index, the index number, a colon and the element value.

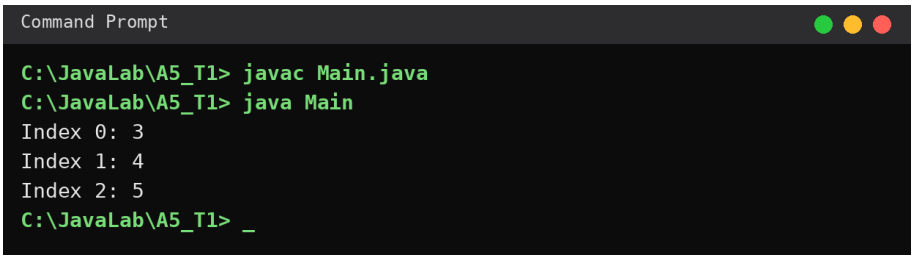
Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 5 - Task 1

public class Main {
    public static void main(String[] args) {
        int[] arr = {3, 4, 5};

        for (int i = 0; i < arr.length; i++) {
            System.out.println("Index " + i + ": " + arr[i]);
        }
    }
}
```

Output:



Task 2: Create an int array with the values 12, 45, 7, 33 and 21. Use a loop to find the sum and the largest value. Print Sum: followed by the sum, then Largest: followed by the largest value.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 5 - Task 2

public class Main {
    public static void main(String[] args) {
        int[] arr = {12, 45, 7, 33, 21};
        int sum = 0;
        int largest = arr[0];

        for (int i = 0; i < arr.length; i++) {
            sum += arr[i];
            if (arr[i] > largest) {
                largest = arr[i];
            }
        }
        System.out.println("Sum: " + sum);
        System.out.println("Largest: " + largest);
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A5_T2> javac Main.java
C:\JavaLab\A5_T2> java Main
Sum: 118
Largest: 45
C:\JavaLab\A5_T2> _
```

Task 3: Create an int array with the values 5, 12, 2, 14 and 7. Use a loop to count the values greater than 10. Print Count greater than 10: followed by the count.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 5 - Task 3

public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 12, 2, 14, 7};
        int count = 0;

        for (int i = 0; i < arr.length; i++) {
            if (arr[i] > 10) {
                count++;
            }
        }
        System.out.println("Count greater than 10: " + count);
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A5_T3> javac Main.java
C:\JavaLab\A5_T3> java Main
Count greater than 10: 2
C:\JavaLab\A5_T3> _
```

Assignment 6: Arrays: Reverse and Duplicates

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Print an array in reverse order, find the smallest value, and check an array for repeated values.*

Task 1: Create an int array with the values 10, 20, 30, 40 and 50. Print the values in reverse order, one value on each line, starting with the last value.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 6 - Task 1

public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 20, 30, 40, 50};

        for (int i = arr.length - 1; i >= 0; i--) {
            System.out.println(arr[i]);
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A6_T1> javac Main.java
C:\JavaLab\A6_T1> java Main
50
40
30
20
10
C:\JavaLab\A6_T1> _
```

Task 2: Create an int array with the values 2, 5, 1, 8, 3 and 4. Use a loop to find the smallest value and its index. Print Smallest: followed by the value, then the words at index and the index.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 6 - Task 2

public class Main {
    public static void main(String[] args) {
        int[] arr = {2, 5, 1, 8, 3, 4};
        int smallest = arr[0];
        int index = 0;

        for (int i = 1; i < arr.length; i++) {
            if (arr[i] < smallest) {
                smallest = arr[i];
                index = i;
            }
        }
        System.out.println("Smallest: " + smallest + " at index " + index);
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A6_T2> javac Main.java
C:\JavaLab\A6_T2> java Main
Smallest: 1 at index 2
C:\JavaLab\A6_T2> _
```

Task 3: Create an int array with the values 4, 9, 2, 9 and 7. Use nested loops to check for a repeated value. Print Repeated value: followed by it, or print No repeated value if none is repeated.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 6 - Task 3

public class Main {
    public static void main(String[] args) {
        int[] arr = {4, 9, 2, 9, 7};
        boolean found = false;

        for (int i = 0; i < arr.length && !found; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] == arr[j]) {
                    System.out.println("Repeated value: " + arr[i]);
                    found = true;
                    break;
                }
            }
        }
        if (!found) {
            System.out.println("No repeated value");
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A6_T3> javac Main.java
C:\JavaLab\A6_T3> java Main
Repeated value: 9
C:\JavaLab\A6_T3> _
```

Assignment 7: Counting Steps and Complexity

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Count the steps a program takes, and write its time complexity as a comment.*

Task 1: Use one loop that runs from 1 to 10, and count its steps. Above the loop, write a comment with its time complexity. Print Steps: followed by the number of steps.

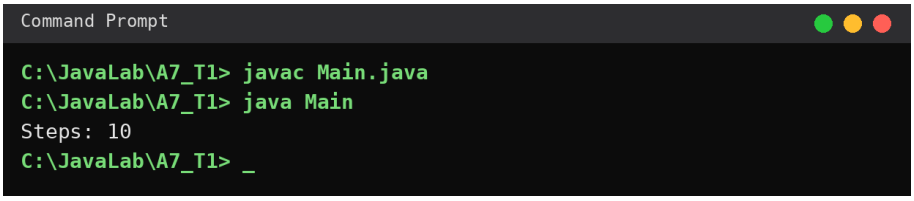
Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 7 - Task 1

public class Main {
    public static void main(String[] args) {
        int steps = 0;

        // Time complexity: O(n) - the loop runs n times
        for (int i = 1; i <= 10; i++) {
            steps++;
        }
        System.out.println("Steps: " + steps);
    }
}
```

Output:



Task 2: Use two loops, one inside the other, that both run from 1 to 10. Count how many times the inner loop body runs. Above the loops, write a comment with their time complexity. Print Steps: followed by the total.

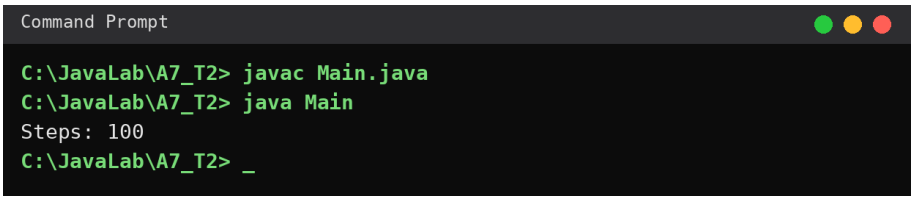
Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 7 - Task 2

public class Main {
    public static void main(String[] args) {
        int steps = 0;

        // Time complexity: O(n^2) - the inner loop runs n times for each of n outer passes
        for (int i = 1; i <= 10; i++) {
            for (int j = 1; j <= 10; j++) {
                steps++;
            }
        }
        System.out.println("Steps: " + steps);
    }
}
```

Output:



Task 3: Store 1024 in an int named n. Use a while loop that halves n until n is 1, and count the divisions. Above the loop, write a comment with its time complexity. Print Halving steps: followed by the count.

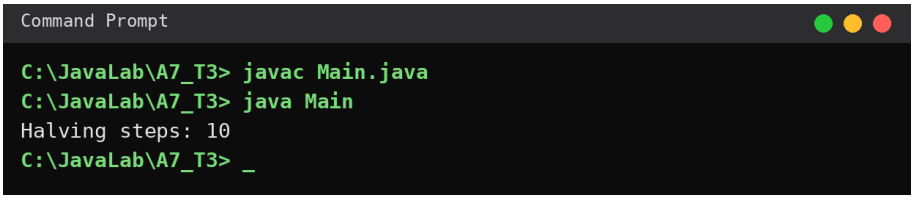
Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 7 - Task 3

public class Main {
    public static void main(String[] args) {
        int n = 1024;
        int steps = 0;

        // Time complexity: O(log n) - n is halved on every pass
        while (n > 1) {
            n = n / 2;
            steps++;
        }
        System.out.println("Halving steps: " + steps);
    }
}
```

Output:



Assignment 8: Matrices: Transpose and Row Sums

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Create two-dimensional arrays, visit them row by row, and build a new matrix from one.*

Task 1: Create a matrix named m with 3 rows and 4 columns. Fill it with the values 1 to 12, row by row, using nested loops. Print the matrix one row per line, with a space after each number.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 8 - Task 1

public class Main {
    public static void main(String[] args) {
        int[][] m = new int[3][4];
        int value = 1;

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 4; j++) {
                m[i][j] = value++;
            }
        }
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 4; j++) {
                System.out.print(m[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A8_T1> javac Main.java
C:\JavaLab\A8_T1> java Main
1 2 3 4
5 6 7 8
9 10 11 12
C:\JavaLab\A8_T1> _
```

Task 2: Create the 3 by 3 matrix with the rows 1 2 3, 4 5 6 and 7 8 9. Create its transpose (rows and columns swapped). Print the transpose one row per line, with a space after each number.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 8 - Task 2

public class Main {
    public static void main(String[] args) {
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
        int[][] t = new int[3][3];

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                t[j][i] = m[i][j];      // rows and columns swapped
            }
        }
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(t[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A8_T2> javac Main.java
C:\JavaLab\A8_T2> java Main
1 4 7
2 5 8
3 6 9
C:\JavaLab\A8_T2> _
```

Task 3: Use the 3 by 3 matrix with the rows 1 2 3, 4 5 6 and 7 8 9. Print the sum of each row, one line per row, in the form Row 1 sum: followed by the total.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 8 - Task 3

public class Main {
    public static void main(String[] args) {
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};

        for (int i = 0; i < 3; i++) {
            int sum = 0;
            for (int j = 0; j < 3; j++) {
                sum += m[i][j];
            }
            System.out.println("Row " + (i + 1) + " sum: " + sum);
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A8_T3> javac Main.java
C:\JavaLab\A8_T3> java Main
Row 1 sum: 6
Row 2 sum: 15
Row 3 sum: 24
C:\JavaLab\A8_T3> _
```

Assignment 9: Matrices: Rows and Columns

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Visit the rows and columns of a matrix with nested loops to print, add and count its values.*

Task 1: Use the 3 by 3 matrix with rows 1 2 3, 4 5 6 and 7 8 9. Print First row: and its values, then Last row: and its values, with a space between values.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 9 - Task 1

public class Main {
    public static void main(String[] args) {
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};

        System.out.print("First row: ");
        for (int j = 0; j < 3; j++) {
            System.out.print(m[0][j] + " ");
        }
        System.out.println();

        System.out.print("Last row: ");
        for (int j = 0; j < 3; j++) {
            System.out.print(m[2][j] + " ");
        }
        System.out.println();
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A9_T1> javac Main.java
C:\JavaLab\A9_T1> java Main
First row: 1 2 3
Last row: 7 8 9
C:\JavaLab\A9_T1> _
```

Task 2: Use the 3 by 3 matrix with the rows 1 2 3, 4 5 6 and 7 8 9. Print the sum of each column, one line per column, in the form Column 1 sum: followed by the total.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 9 - Task 2

public class Main {
    public static void main(String[] args) {
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};

        for (int j = 0; j < 3; j++) {
            int sum = 0;
            for (int i = 0; i < 3; i++) {
                sum += m[i][j];
            }
            System.out.println("Column " + (j + 1) + " sum: " + sum);
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A9_T2> javac Main.java
C:\JavaLab\A9_T2> java Main
Column 1 sum: 12
Column 2 sum: 15
Column 3 sum: 18
C:\JavaLab\A9_T2> _
```

Task 3: Use the 3 by 3 matrix with the rows 1 2 3, 4 5 6 and 7 8 9. Count the values greater than 5 and print Values greater than 5: followed by the count.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 9 - Task 3

public class Main {
    public static void main(String[] args) {
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
        int count = 0;

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                if (m[i][j] > 5) {
                    count++;
                }
            }
        }
        System.out.println("Values greater than 5: " + count);
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A9_T3> javac Main.java
C:\JavaLab\A9_T3> java Main
Values greater than 5: 4
C:\JavaLab\A9_T3> _
```

Assignment 10: Switch and ArrayList

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Choose a branch with a switch statement, and store and change a list of numbers in an ArrayList.*

Task 1: Store 3 in an int named day. Use a switch statement to print the name of the day: 1 is Monday, 2 is Tuesday and 3 is Wednesday. For any other number, print Invalid day.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 10 - Task 1

public class Main {
    public static void main(String[] args) {
        int day = 3;

        switch (day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            default:
                System.out.println("Invalid day");
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A10_T1> javac Main.java
C:\JavaLab\A10_T1> java Main
Wednesday
C:\JavaLab\A10_T1> _
```

Task 2: Create an ArrayList of Integer named list. Add the values 10, 20, 30 and 40. Print the whole list. Then print Size: followed by its size, and Item at index 2: followed by the value at index 2.

Program (Main.java):

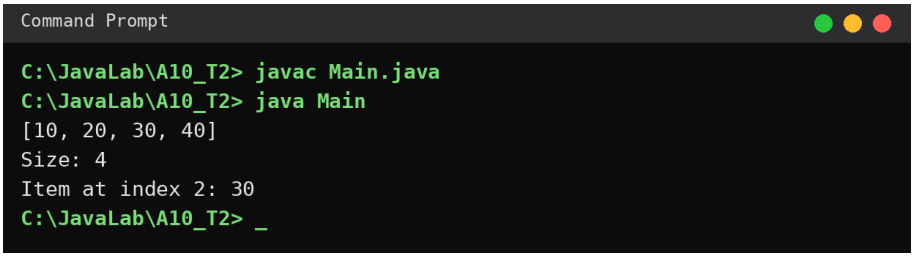
```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 10 - Task 2

import java.util.ArrayList;

public class Main {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);

        System.out.println(list);
        System.out.println("Size: " + list.size());
        System.out.println("Item at index 2: " + list.get(2));
    }
}
```

Output:



Task 3: Create an ArrayList of Integer named list with the values 10, 20, 30 and 40. Use set to change index 1 to 25, and remove to delete index 3. Print the list and Size: followed by its size.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 10 - Task 3

import java.util.ArrayList;

public class Main {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);

        list.set(1, 25);     // change index 1 to 25
        list.remove(3);      // delete index 3

        System.out.println(list);
        System.out.println("Size: " + list.size());
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A10_T3> javac Main.java
C:\JavaLab\A10_T3> java Main
[10, 25, 30]
Size: 3
C:\JavaLab\A10_T3> _
```

Assignment 11: Bubble Sort and Selection Sort

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Sort small arrays with bubble sort and selection sort, and count the swaps.*

Task 1: Create the int array 5, 1, 4, 2, 8. Sort it in increasing order with bubble sort. Print Sorted: followed by the sorted values in square brackets, separated by commas.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 11 - Task 1

import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 1, 4, 2, 8};
        int n = arr.length;

        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - 1 - i; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A11_T1> javac Main.java
C:\JavaLab\A11_T1> java Main
Sorted: [1, 2, 4, 5, 8]
C:\JavaLab\A11_T1> _
```

Task 2: Create the int array 64, 25, 12, 22, 11. Sort it in increasing order with selection sort. Print Sorted: followed by the sorted values in square brackets, separated by commas.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 11 - Task 2

import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        int[] arr = {64, 25, 12, 22, 11};
        int n = arr.length;

        for (int i = 0; i < n - 1; i++) {
            int min = i;
            for (int j = i + 1; j < n; j++) {
                if (arr[j] < arr[min]) {
                    min = j;
                }
            }
            int temp = arr[min];
            arr[min] = arr[i];
            arr[i] = temp;
        }
        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A11_T2> javac Main.java
C:\JavaLab\A11_T2> java Main
Sorted: [11, 12, 22, 25, 64]
C:\JavaLab\A11_T2> _
```

Task 3: Use the array 5, 1, 4, 2, 8 and sort it with bubble sort. Count the swaps (each time two values change places). Print Swaps: followed by the count.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 11 - Task 3

public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 1, 4, 2, 8};
        int n = arr.length;
        int swaps = 0;

        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - 1 - i; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                    swaps++;
                }
            }
        }
        System.out.println("Swaps: " + swaps);
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A11_T3> javac Main.java
C:\JavaLab\A11_T3> java Main
Swaps: 4
C:\JavaLab\A11_T3> _
```

Assignment 12: Insertion Sort

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Sort small arrays with insertion sort and count how many values are shifted.*

Task 1: Create the int array 7, 3, 5, 1. Sort it in increasing order with insertion sort. Print Sorted: followed by the sorted values in square brackets, separated by commas.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 12 - Task 1

import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        int[] arr = {7, 3, 5, 1};

        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];     // shift right
                j--;
            }
            arr[j + 1] = key;
        }
        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A12_T1> javac Main.java
C:\JavaLab\A12_T1> java Main
Sorted: [1, 3, 5, 7]
C:\JavaLab\A12_T1> _
```

Task 2: Create the int array 14, 3, 9, 21, 6. Sort it in increasing order with insertion sort. Print Sorted: followed by the sorted values in square brackets, separated by commas.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 12 - Task 2

import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        int[] arr = {14, 3, 9, 21, 6};

        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];     // shift right
                j--;
            }
            arr[j + 1] = key;
        }
        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A12_T2> javac Main.java
C:\JavaLab\A12_T2> java Main
Sorted: [3, 6, 9, 14, 21]
C:\JavaLab\A12_T2> _
```

Task 3: Create the int array 6, 5, 4, 3, 2, 1. Sort it in increasing order with insertion sort. Count the shifts (each time one value moves one place to the right). Print Shifts: followed by the count.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 12 - Task 3

public class Main {
    public static void main(String[] args) {
        int[] arr = {6, 5, 4, 3, 2, 1};
        int shifts = 0;

        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                shifts++;
                j--;
            }
            arr[j + 1] = key;
        }
        System.out.println("Shifts: " + shifts);
    }
}
```

Output:



Assignment 13: Strings: Methods and Palindromes

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Use common String methods, reverse a word, check a palindrome and print substrings.*

Task 1: Use the word Programming. Print Length: followed by its length, Position of g: followed by the position of the first g (counting from 0), and Upper case: followed by the word in capital letters, using toUpperCase().

Program (Main.java):

```
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 13 - Task 1

public class Main {
    public static void main(String[] args) {
        String word = "Programming";

        System.out.println("Length: " + word.length());
        System.out.println("Position of g: " + word.indexOf('g'));
        System.out.println("Upper case: " + word.toUpperCase());
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A13_T1> javac Main.java
C:\JavaLab\A13_T1> java Main
Length: 11
Position of g: 3
Upper case: PROGRAMMING
C:\JavaLab\A13_T1> _
```

Task 2: Reverse the word Java. Build the reversed word one letter at a time, using charAt and + in a loop. Print Reversed: followed by the result.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 13 - Task 2

public class Main {
    public static void main(String[] args) {
        String word = "Java";
        String rev = "";

        for (int i = word.length() - 1; i >= 0; i--) {
            rev = rev + word.charAt(i);
        }
        System.out.println("Reversed: " + rev);
    }
}
```

Output:



Task 3: Use the word racecar. Check whether it is a palindrome, which means it reads the same from both ends. Print the word followed by is a palindrome, or followed by is not a palindrome.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 13 - Task 3

public class Main {
    public static void main(String[] args) {
        String word = "racecar";
        boolean palindrome = true;

        for (int i = 0; i < word.length() / 2; i++) {
            if (word.charAt(i) != word.charAt(word.length() - 1 - i)) {
                palindrome = false;
                break;
            }
        }
        if (palindrome) {
            System.out.println(word + " is a palindrome");
        } else {
            System.out.println(word + " is not a palindrome");
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A13_T3> javac Main.java
C:\JavaLab\A13_T3> java Main
racecar is a palindrome
C:\JavaLab\A13_T3> _
```

Task 4: Use the word abc. Print every substring of it, one on each line, using substring(). For each starting letter from left to right, print its parts from shortest to longest.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 13 - Task 4

public class Main {
    public static void main(String[] args) {
        String word = "abc";

        for (int i = 0; i < word.length(); i++) {
            for (int j = i + 1; j <= word.length(); j++) {
                System.out.println(word.substring(i, j));
            }
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A13_T4> javac Main.java
C:\JavaLab\A13_T4> java Main
a
ab
abc
b
bc
c
C:\JavaLab\A13_T4> _
```

Assignment 14: Characters: Counting and Positions

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Use charAt to count vowels and a letter, and print each letter with its position.*

Task 1: Store programming in a String named s. Count its vowels (a, e, i, o and u). Print Vowels: followed by the count.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 14 - Task 1

public class Main {
    public static void main(String[] args) {
        String s = "programming";
        int count = 0;

        for (int i = 0; i < s.length(); i++) {
            char c = s.charAt(i);
            if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') {
                count++;
            }
        }
        System.out.println("Vowels: " + count);
    }
}
```

Output:



Task 2: Store programming in a String named s. Count how many times the letter g appears. Print Count of g: followed by the count.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 14 - Task 2

public class Main {
    public static void main(String[] args) {
        String s = "programming";
        int count = 0;

        for (int i = 0; i < s.length(); i++) {
            if (s.charAt(i) == 'g') {
                count++;
            }
        }
        System.out.println("Count of g: " + count);
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A14_T2> javac Main.java
C:\JavaLab\A14_T2> java Main
Count of g: 2
C:\JavaLab\A14_T2> _
```

Task 3: Store hello in a String named s. Print each letter on its own line as its position, a colon, a space and the letter. Positions start at 0.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 14 - Task 3

public class Main {
    public static void main(String[] args) {
        String s = "hello";

        for (int i = 0; i < s.length(); i++) {
            System.out.println(i + ": " + s.charAt(i));
        }
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A14_T3> javac Main.java
C:\JavaLab\A14_T3> java Main
0: h
1: e
2: l
3: l
4: o
C:\JavaLab\A14_T3> _
```

Assignment 15: StringBuilder Methods

Name: *Ujjwal Mandal* ***Student ID:*** *BCS2026443*

Aim: *Build and change text with StringBuilder, using append, insert and deleteCharAt.*

Task 1: Create a StringBuilder that starts with the text Hello. Append the text " World" to it. Print the StringBuilder, then print Length: followed by its length.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 15 - Task 1

public class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Hello");
        sb.append(" World");

        System.out.println(sb);
        System.out.println("Length: " + sb.length());
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A15_T1> javac Main.java
C:\JavaLab\A15_T1> java Main
Hello World
Length: 11
C:\JavaLab\A15_T1> _
```

Task 2: Use a StringBuilder. Append each number from 1 to 5, followed by a comma. Then remove the last comma with deleteCharAt. Print the result.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 15 - Task 2

public class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder();

        for (int i = 1; i <= 5; i++) {
            sb.append(i).append(",");
        }
        sb.deleteCharAt(sb.length() - 1);   // remove the last comma
        System.out.println(sb);
    }
}
```

Output:

```
Command Prompt

C:\JavaLab\A15_T2> javac Main.java
C:\JavaLab\A15_T2> java Main
1,2,3,4,5
C:\JavaLab\A15_T2> _
```

Task 3: Create a StringBuilder that starts with the text Java. Use insert to add the text Hello followed by a space at index 0. Print the StringBuilder.

Program (Main.java):

```java
// Name: Ujjwal Mandal
// Student ID: BCS2026443
// Assignment 15 - Task 3

public class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Java");
        sb.insert(0, "Hello ");

        System.out.println(sb);
    }
}
```

Output:

